# Battery-operated/electronic toys: HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for battery-operated/electronic toys in Children's Toys, Arts, and Crafts, evaluated on element MIGRATION from scraped-off toy material (EN 71-3 Category III, mg/kg of material): a migration ceiling, not a total-content ceiling. 8 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; strictest applicable sovereign maximum in the row's native basis, disclosed read-across where the exact cell is unregulated; never occurrence-derived). Category 21 publishes on the element-migration basis only, pooling the EU, US and Mexican migration instruments, with each row pinned to the EN 71-3 material category or categories its scope actually spans.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 23,000 | government maximum level |
| Cadmium (Cd) | Tier 1 | 17,000 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 25,000 | government maximum level |

| | | | |
|---|---|---|---|
| Inorganic arsenic (iAs) | Tier 1 | n/m | not material in this matrix |
| Mercury, total (tHg) | Tier 1 | 60,000 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | n/m | not material in this matrix |
| Nickel (Ni) | Tier 2 | 930,000 | government maximum level |
| Tin (Sn) | Tier 2 | 180,000,000 | screening ceiling (European Union) |
| Aluminium (Al) | Tier 2 | 28,130,000 | government maximum level |
| Total chromium (Cr) | Tier 2 | 60,000 | screening ceiling (US FDA) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 53 | government maximum level |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 23,000 | government maximum level |
| Cadmium (Cd) | 17,000 | government maximum level |
| Total arsenic (tAs) | 25,000 | government maximum level |
| Inorganic arsenic (iAs) | n/m | No sovereign toy-safety instrument sets an inorganic-arsenic limit. Directive 2009/48/EC Annex II Part III point 13, EN 71-3, ISO 8124-3:2020 Table 1, ASTM F963-17 4.3.5 and NOM-252-SSA1-2011 Table 1 all regulate TOTAL arsenic migration from toy material; inorganic arsenic is a dietary speciation with no article-migration counterpart. Arsenic in an accessible housing, keypad, lens, cable sheath or plated contact is bounded by the published total-arsenic migration ceiling by identity (iAs <= tAs), so a separate inorganic-arsenic ceiling would add no control. |

| | | |
|---|---|---|
| Mercury, total (tHg) | 60,000 | government maximum level |
| Methylmercury (MeHg) | n/m | Methylmercury is produced by microbial methylation in aquatic systems and reaches humans by diet. an accessible housing, keypad, lens, cable sheath or plated contact carries no methylation pathway and no dietary route, and no sovereign toy instrument sets a methylmercury limit: every toy mercury limit (2009/48/EC point 13, ASTM F963-17 4.3.5, ISO 8124-3:2020, NOM-252-SSA1-2011, Canada SOR/2011-17 and SOR/2016-193) is on total mercury. Any organomercury fraction is bounded by the published total-mercury migration ceiling by identity (MeHg <= tHg). |
| Nickel (Ni) | 930,000 | government maximum level |
| Tin (Sn) | 180,000,000 | government maximum level |
| Aluminium (Al) | 28,130,000 | government maximum level |
| Total chromium (Cr) | 60,000 | government maximum level |
| Hexavalent chromium (Cr-VI) | 53 | government maximum level |

## 4 | Cite this standard

Pendergrass, K. (2026). *Battery-operated/electronic toys: HMTc standard*. Institute of Contaminant Standards.

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### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/toys-battery-electronic. Primary legal instruments and expert-body assessments are

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named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.